M1S2 | OOP

Deep

Class is ...

an object’s draft which defines its attributes and methods

OOP Principles

Inheritance

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data and methods related to an object are combined in one place (class)

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Abstraction

allows you to highlight the general characteristics and functionality of objects or systems, ignoring the implementation details

OOP Pros & Cons

✅Modularity and Code Reuse

allows you to create classes that can serve as reusable modules

✅Improved Separation of Concerns

makes it easy to identify which part of the code is responsible for a specific functionality

❌Tendency to Be Confusing

class hierarchies often lead to multiple method overridings

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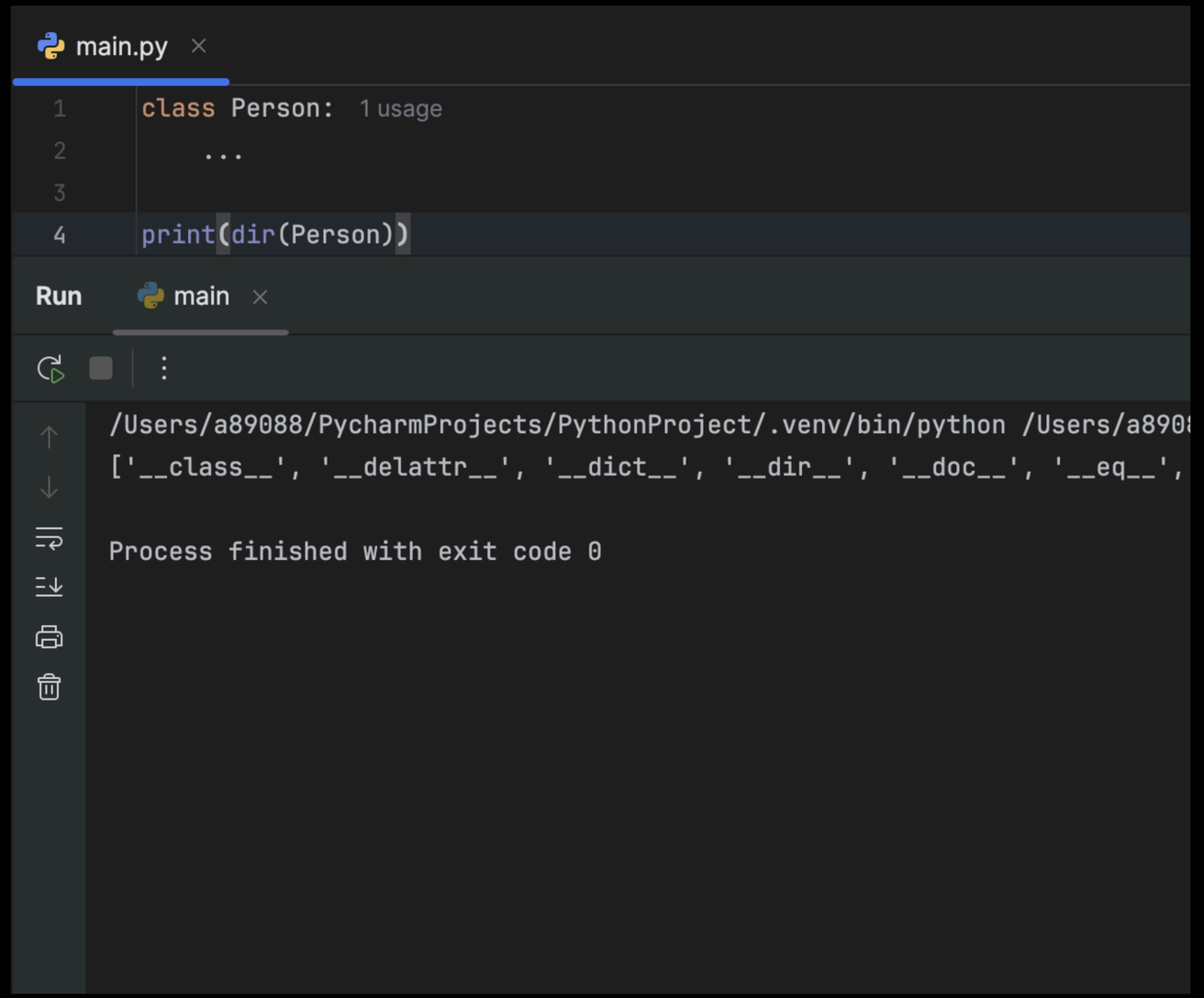
Lang Construction

Dunder(Magic) Methods



__init__, __str__, __repr__, __len__, __getitem__, __iter__, ...

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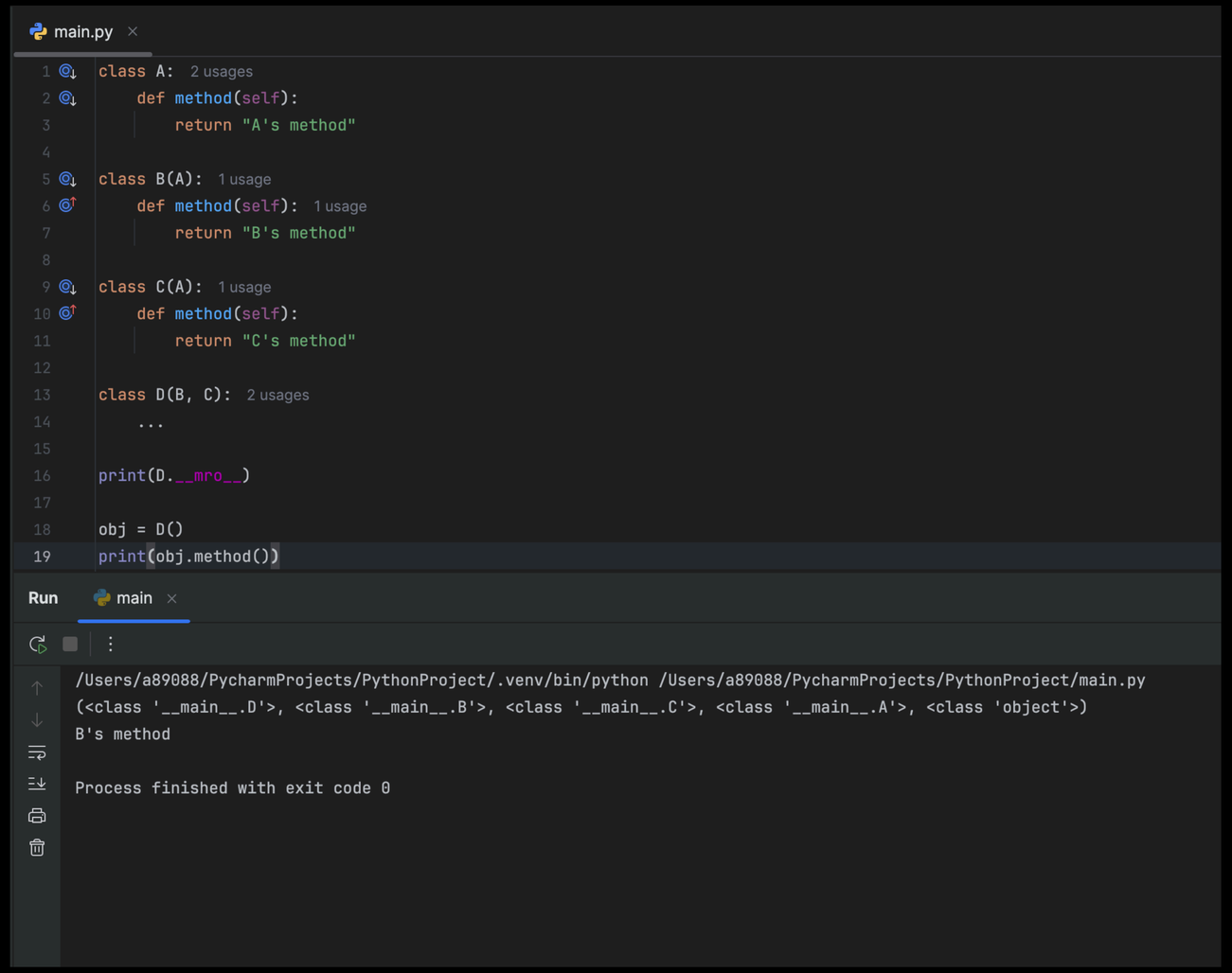
MRO(Method Resolution Order)

Multiple Inheritance



algorithm for searching methods and attributes in inheritance hierarchy

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SOLID💀

Single Responsibility

module must be responsible for one and only one actor

Open-Closed

software entities should be open for extension, but closed for modification

Liskov Substitution

functions that use a base type should be able to use subtypes of the base type without knowing about it

Interface Segregation

multiple client-specific interfaces are better than one general-purpose interface

Dependency Inversion

dependency on Abstractions. No dependency on implementation